Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Physics-Grounded Real2Sim Farm Simulator for Crop Manipulation with 3D Gaussian Splatting and Policy/Motion Planning Deployment**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials			
	Medium Category	Small Category	Medium Category	Small Category
	Robotics/Automation Machinery	*Robot Control and Intelligence Technologies*		
Concept of Proposed Technology	**Core Concept** This proposal establishes an end-to-end pipeline that (i) builds a Real2Sim farm digital twin from indoor and outdoor videos, (ii) learns manipulation policies via physics-grounded simulation and reinforcement learning (RL), and (iii) transfers the combined policy and motion-planning stack to real robots for quantified field validation. Agricultural manipulation is dominated by lighting variation, occlusion, crop deformability, and contact-rich interactions including cutting, which makes iterative real-world training costly and risk-prone. A closed-loop workflow—video-to-digital-twin → physics/RL policy learning → sim-to-real deployment and evaluation—is therefore essential. ● (Stage 1: Digital Twin / Assetization) The proposer formalizes capture settings for robust 3D reconstruction (camera selection, field of view, acquisition speed, and trajectories) and adapts 3D Gaussian Splatting (3DGS) and related 2D-to-3D learning methods to the crop			

	domain, moving beyond rigid-body assumptions to enable crop and farm environment assetization. ● (Stage 2: Physics + RL) With collaborating partners, the project assigns and refines per-asset physical properties (stiffness, damping, friction) and builds numerically stable models for deformable objects such as stems, leaves, and fruits. Based on these models, RL policies are trained to achieve robust grasping and manipulation across diverse physical conditions. ● (Stage 3: Sim-to-Real) With a Korean SME partner, the robot manipulator and end-effector are faithfully modeled in the simulator, including physical parameters required for contact and cutting. The learned policies and motion-planning components are deployed to real robots and evaluated using quantitative KPIs such as success rate, damage rate, cycle time, and force/safety constraint violations, while systematically reducing the sim-to-real gap. **Key Functions and Potential Applications** **Key Functions** ● A video-driven digital-twin and asset pipeline for rapid updates of farms and crops in simulation ● Observation-based physical parameterization, including Gaussian-based object separation for deformables, initialization (e.g., rod-based stems, mass-spring fruits), and refinement via priors and probe actions ● Stable physics simulation for sustained manipulation using volume preservation constraints for near-incompressible deformables, damping, compliant contact modeling, and numerical optimization ● RL policies that generalize across physical and environmental variability ● Integrated deployment of policies and motion planning for collision-aware execution and KPI-based field validation ● A structured sim-to-real gap reduction framework via domain randomization, parameter tuning, and integration of teleoperation data **Potential Applications** ● Fruit harvesting, flower thinning, pruning, stem/branch cutting, contact-based crop handling, and broader greenhouse and orchard manipulation tasks.
Rationale for Joint R&D Project	**Critical Technical Barriers** Agricultural manipulation is inherently high-risk and high-variance: robots must interact physically with crops (grasping, pulling, twisting, cutting) under severe occlusions, dynamic lighting, growth-driven geometry changes, and large object-level physical variability. In this domain, isolated advances in perception or control often plateau at lab demonstrations, because performance collapses when exposed to real farm conditions. To achieve meaningful progression, the project must be designed as a single chain 1) Real2Sim digital twin construction 2) Physics-grounded learning for deformable crop interaction 3) Deployment with KPI-based field validation.

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	Why Global Collaboration? 1) Why Real2Sim digital twin research is needed. Policy learning and motion planning are ultimately bounded by the fidelity of simulation inputs—farm and crop assets. Because farms exhibit extreme lighting variation, occlusion, and continuous crop changes, generic 3D reconstruction is insufficient for manipulation-grade realism. The proposer brings established know-how in building digital twins from indoor/outdoor farm videos and operationalizing them for simulation. By systematizing capture protocols (camera choice, FoV, acquisition speed, trajectories) and advancing 3D Gaussian Splatting (3DGS)-based rendering/assetization, the proposer can deliver a Real2Sim pipeline that is faithful enough to support transferable learning, which is the critical bottleneck for the downstream stack. 2) Why deformable physics and RL must be integrated Unlike many domains, agricultural automation cannot avoid contact with deformable crops; rendering-only simulation does not transfer. Fruits and stems exhibit near-incompressible behavior, complex contact dynamics, and object-level variability that rigid-body simulators handle poorly. A learnable environment for deformable manipulation requires, simultaneously, (i) high-quality 3D observations and digital representation, (ii) deformable physical modeling with numerical stabilization, and (iii) reinforcement learning and validation pipelines. Building all of these capabilities in-house is typically infeasible within realistic time and resource constraints. An international joint R&D structure enables a division of labor that connects these specialized capabilities into a cohesive, executable pipeline. 3) Why real-world deployment and field validation must be part of the project TRL advancement requires more than simulation success; it requires field KPIs such as task success rate, crop damage rate, cycle time, and safety/force constraint violations. Without real-robot integration, end-effector realism, farm testbeds, and repeated on-site iterations, results remain unverified and do not translate into deployable performance. A Korean SME partner with commercialization experience in agricultural robotics provides the practical bridge: deep-learning perception, task-optimized end-effectors, controlled crop testbeds (e.g., strawberries, cucumbers, tomatoes), and access to deployable customer sites. This enables the indispensable deploy–measure–improve loop that converts simulation outcomes into field-proven performance, making TRL 6 outcomes achievable.
Key Words	1. Real2Sim (Video-to-Simulation) 2. 3D Gaussian Splatting (3DGS) 3. Deformable Crop Physics 4. Physics-Grounded Reinforcement Learning 5. Sim-to-Real Deployment & Validation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (6)

R&D Objectives

R&D Objectives

The objective of this project is to (i) build a Real2Sim simulator for agricultural manipulation, (ii) obtain physics-grounded interaction models and RL policies for deformable crops, and (iii) deploy the policy and motion/path planning stack on real robots and validate performance through field KPIs.

Detailed Objectives

Establish a 3DGS-Based Farm Simulation Asset Pipeline

1) Standardize Real2Sim data acquisition from farm videos
   - Define capture protocols for reliable 3D reconstruction in indoor/outdoor farms (camera/FoV, capture speed, trajectories, lighting conditions, calibration).
   - Construct multi-view datasets covering crops, structures, and workspaces with QA criteria.
2) Digital twin reconstruction and rendering with 3D Gaussian Splatting (3DGS)
   - Build a reconstruction pipeline for scenes including farm structures and crops.
   - Develop 3DGS training/optimization procedures to mitigate crop-domain challenges (rigid-body assumptions, occlusions, lighting shifts, growth-driven changes).
   - (Optional) Integrate auxiliary pipelines (e.g., segmentation-assisted reconstruction) for crop element separation and refinement.
3) Assetization and interfaces for simulator integration
   - Package assets into visual, semantic, and interaction layers (semantic labels, collision proxies, reference frames, actionable regions).
   - Define conversion specifications from digital twin outputs to simulator-ready assets (data formats, metadata schema, versioning), enabling reusability.
4) Asset quality verification
   - Establish validation procedures for reconstruction fidelity and repeatability (e.g., multi-view consistency, geometric/visual alignment metrics).
   - Build update workflows to refresh assets efficiently as farms change (growth, layout updates).

Learn and Validate Physics-Grounded Interaction and RL Policies

1) Deformable crop simulation with object-level physical alignment
   - Use Gaussian-based 3D observations to automatically classify stems (rod) and fruits (mass-spring), reflecting near-incompressible behavior via:
     - non-uniform bending stiffness (EI) rod models for stems,
     - volume-preserving constraints in mass-spring models for fruits.
   - Combine image/crop-informed physical priors with ≤5-second probe motions to:
     - segment objects into 2–3 physical regions,

- iteratively calibrate contact/friction/damping/stiffness at the instance level.

2) Numerically stable interaction simulation for sustained manipulation
   - Ensure stable simulation that supports >10 seconds of continuous manipulation under:
     - physical variability (stiffness/friction/damping ±30%),
     - pose uncertainty (±2–3 cm).
   - Apply stabilization techniques (volume preservation, damping, compliant contact models) to prevent blow-up or collapse.
3) Contact-centric RL policy acquisition and benchmarking
   - Train RL policies for robust grasping/manipulation across varying physical conditions.
   - Target performance: ≥90% grasp success rate.
   - Build benchmark scenarios to evaluate generalization (physics variations, pose uncertainty, occlusions, lighting changes).

Deploy on Real Robots with Motion Planning and Field KPI Evaluation

1) Robot and end-effector assetization inside the simulator
   - Assetize the existing manipulator and end-effector, incorporating actuation/joint constraints, sensing, and contact models.
   - Assign end-effector physical parameters (friction, stiffness/compliance) required for contact/cutting and establish alignment procedures between sim and hardware.
2) Integrated deployment stack: policy + motion/path planning
   - Deploy simulation-trained policies onto the real robot control stack.
   - Integrate collision-aware motion planning (approach poses, task sequencing, safety constraints) for real-time execution.
3) Sim-to-real gap reduction and field validation
   - Develop sim-to-real mitigation methods (parameter calibration, domain randomization, residual correction) driven by field data.
   - Combine teleoperation-derived learning data with RL policies to improve field success probability.
   - Establish KPI-based evaluation and iterative improvement loops in real testbeds (success, damage, cycle time, force/safety violations, etc.).

R&D Target Specifications

Key KPI	Target	Weight	Measurement & Verification
Reprojection Error	**mean/median ≤ 2 px** *(recommended)*	20%	Reconstruct a 3D scene from multi-view captures, project 3D back to each camera view, and compute pixel-level errors against observed 2D points.
Policy Success Rate (Task/Grasp)	**≥ 90%** *(physics ±30%, pose ±2–3 cm)*	25%	Run **N episodes** with randomized physical parameters (stiffness/friction/damping ±30%) and initial pose perturbations (±2–3 cm).
Simulation Stability	**max penetration ≤ 5 mm @ ≥30 fps**	15%	Log penetration depth per frame during contact-rich scenarios. Execute ≥10 s continuous manipulation at ≥30 fps, and verify max penetration depth stays ≤5 mm (

Field Task Success Rate	**≥ 80%** *(tomato harvesting unit task)*	25%	Conduct **N trials** in the field. Evaluate success using a pre-agreed checklist (perception→approach→grasp→detach→place), including damage/quality thresholds
Cycle Time per Unit Task	**≤ 15 s/task** *(tomato harvesting unit task)*	15%	Define start/end events (e.g., "approach start"→"harvest complete"). Compute cycle time from field timestamps and report mean/median

R&D Plan

Research Approach & Strategy

Integrated pipeline: Data → Simulation → Learning → Motion/Path Planning → Field Validation

The project is structured to ensure TRL advancement by connecting Real2Sim digital twin construction, physics-grounded interaction simulation, RL policy learning, integration with motion/path planning and safety constraints, and KPI-driven field validation in a single loop.

- Proposer: establish high-fidelity Real2Sim digital twins using 3DGS, including capture protocols, alignment, assetization, and simulator-ready packaging.
- Korean SME partner: adopt an observation–physics–learning strategy; constrain RL to contact-centric segments (grasp/hold) to achieve robustness under physics variability and pose errors.
- Korean SME partner: deploy policy + motion planning as an executable stack, validate in real farms with KPIs, and iteratively reduce the sim-to-real gap.

Key Tasks & Milestones (Year/Quarter)

Year 1: Real2Sim pipeline + stable deformable simulation foundation

Year 2: instance-level physics alignment + learning/validation pipeline integration

Year 3: deployment with motion planning + KPI-based field PoC and TRL evidence

Methodology

- 3DGS Real2Sim: capture → pose/alignment → 3DGS optimization → semantic/collision proxy extraction → simulator asset packaging
- Deformable physics & RL: observation-based classification → probe-assisted calibration → numerical stabilization → contact-centric RL with domain randomization
- Deployment & validation: integrated policy+planning execution → sim-to-real gap measurement → iterative KPI-driven field improvement

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	1) Real2Sim-based farm digital twin construction ● Establish a multi-view video–driven 3D Gaussian Splatting (3DGS) reconstruction and rendering pipeline for indoor and outdoor farm environments, explicitly accounting for occlusions, lighting variation, and crop growth dynamics. 2) Simulator-ready assetization framework ● Define the full assetization stack required to convert digital twin outputs into simulator-importable assets, including coordinate-frame alignment, semantic labeling, collision proxies (collision models), and a metadata schema. 3) Extension to crop-domain representations ● Address the limitations of rigid-body–centric 3D representations by developing data capture, preprocessing, and optimization procedures that reduce omissions and distortions in crop elements (stems, leaves, and fruits).
Expected R&R of Korean SME Partner	1) Technical Contributions and Development Tasks ● Observation-driven physics alignment and stable simulation: Use Gaussian-based 3D observations to automatically classify deformable objects—stems (rod) and fruits (mass-spring)—and develop physics models and numerical stabilization methods that capture near-incompressible behavior. ● Probe-conditioned RL control framework: Build a contact-centric reinforcement learning control framework in which probe actions calibrate instance-level physical properties, enabling stable grasping and holding across diverse physics conditions. ● Scope optimization for learnability and robustness: Constrain RL to the grasping/contact-maintenance segment (rather than exploration or global path planning) to obtain policies robust to physics variability and pose errors. ● Robot (manipulator/EEF) modeling and simulator assetization: Model the real robot system for simulator execution (kinematics, joint constraints, sensors, contact points) and assign end-effector physical properties (e.g., friction, stiffness, compliance) to reproduce task-relevant interactions. ● Integrated execution stack for policy + motion/path planning: Integrate learned policies with motion/path planning—including safety constraints, collision avoidance, approach pose, and task sequencing—so the stack can be executed repeatedly on real robots. ● Sim-to-real gap reduction and field deployment validation: Reduce the sim-to-real gap and improve performance through field-data-driven calibration/parameter tuning and by combining teleoperation-based data accumulation with policy learning.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical Outcomes** 1) Foundational and academic contributions ● Establish a 3DGS-based Real2Sim pipeline—from capture and alignment to reconstruction and assetization—that produces simulator-importable farm digital twins even under occlusion, lighting variation, and crop growth changes across indoor and outdoor environments. ● Deliver a reusable farm asset standard including coordinate frames, semantic labels, collision proxies, and versioning rules, enabling consistent reuse in both research and industrial deployment. ● Package the results into publishable artifacts: capture protocols and quality metrics (e.g., reprojection error, multi-view consistency) alongside digital-twin outputs as datasets/benchmarks, enabling papers, workshops, and open-source release where appropriate. 2) Learning and simulation outcomes ● Build a physics-grounded interaction simulator for deformable crops by combining stem-as-rod and fruit-as-mass-spring modeling with near-incompressible stabilization (volume preservation, damping, contact models). ● Develop a probe-assisted physics calibration + RL control workflow, allowing instance-level property tuning before task execution and training/validation of contact-centric RL policies robust to physics variation and pose errors. ● Provide reusable benchmarks and task packages (grasp/hold/detach), with reproducible evaluation scenarios covering physics distributions, pose error, and occlusion—transferable across robots and follow-on projects. **Commercialization and Field Deployment Outcomes** 1) Field KPI improvements and product/service packaging (Korean SME partner) ● Establish a KPI-driven TRL uplift loop by validating performance using field metrics (task success rate, cycle time, damage rate, stop/intervention rate, safety constraint violations) and by quantifying and reducing the sim-to-real gap through iterative improvement. ● Productize an execution stack that integrates learned policies with motion/path planning and safety constraints, converting one-off demos into repeatable, operations-ready automation modules. ● Enable a scalable adoption and maintenance model by continuously updating digital twins as farms change and re-training/re-validating policies—supporting expansion from PoC to commercial rollout across multiple farms. 2) SME and Industrial/Economic Impact ● Accelerate the domestic ag-robotics ecosystem by providing standardized data, assets, tasks, and evaluation protocols, lowering the barrier for SMEs to scale across crops and operations.

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- Strengthen competitiveness in advanced robotics and manufacturing by demonstrating a credible pathway from deformable-contact research to field-proven TRL advancement and commercialization.

3) Expansion Roadmap

- Crop expansion: Starting from tomato/cucumber/strawberry validation, accumulate reusable digital-twin, physics, and learning assets extensible to structurally similar fruiting vegetables and fruits.
- Task expansion: Extend beyond harvesting into contact-centric operations (thinning, pollination, stationary manipulation, selected pruning), while continuously building task-specific KPI definitions and a growing task library.

6. References